WHITE PAPER · FADAL VMC 4020-906-1 × MILITARY & DEFENSE

Fadal VMC 4020 for Military & Defense

CNC vertical mill for precision milling, drilling, and contouring on prototypes through medium production runs. This paper covers what the machine does, the military & defense parts we produce with it, the alloys we run, the tolerances and finishes military & defense buyers expect, and the documentation package.

Pistol Whiskers · 20" W × 48" L × 24" H workpiece envelope · CNC vertical machining center

Executive summary

Machine: Fadal VMC 4020-906-1 (shop nickname: *Pistol Whiskers*). CNC vertical machining center at B&R Productions, running from our New Waverly, Texas shop under an ISO 9001:2015 quality system.

Application: CNC vertical mill for precision milling, drilling, and contouring on prototypes through medium production runs for Military & Defense customers — Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

Envelope: 20" W × 48" L × 24" H workpiece envelope. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

The Fadal 4020 is the workhorse of American medium-envelope CNC milling — rigid, well-understood, and forgiving with the harder alloys. Pistol Whiskers is our go-to for prototype work, repair parts, and small-to-medium production runs where the part fits inside the envelope. It handles complex pocketing, milled features, and multi-face setups without flinching on the alloys that would chatter marginal equipment.

Full spec table

Model	Fadal VMC 4020-906-1 ("Pistol Whiskers")
Type	3-axis CNC vertical machining center
Workpiece envelope	20" W × 48" L × 24" H (B&R working spec)
Axis travels	40" X × 20" Y × 20–28" Z (Fadal 4020 platform)
Table size	48" × 20"
Spindle taper	CAT 40
Spindle speed	75 – 7,500 RPM standard; 10,000 RPM on HT-series builds
Spindle drive	15 HP with 220 ft-lb torque (HT class)
Rapid traverse	900 IPM (per axis)
Automatic tool changer	21-tool ATC
Control	Fadal CNC-88 / Format 2 (some HT and later builds run Fanuc 0i-MC)
Machine weight	~12,500 lb
Tolerance capability	±0.0005" routine on critical features
Materials	Aluminum, stainless, tool steel, 4140/4340, Inconel, Hastelloy, titanium

Who this serves in Military & Defense

Typical buyer: Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

What's hard about military & defense work: ITAR compliance, security-sensitive prints, program-specific quality plans, tight-tolerance structural components, exotic-alloy selection driven by service environment, and legacy-part sustainment where the OEM no longer supports the design.

Typical Military & Defense parts we produce on Pistol Whiskers

- Structural brackets, mounting plates, and adapters
- Weapon housings and receivers within 20" envelope
- Vehicle armor bracket hardware and mounts
- Sensor housings and control-system enclosures
- Reverse-engineered legacy parts for sustainment programs

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Military & Defense

Standard range: 17-4 PH · 15-5 PH · 13-8 Mo · 4140 pre-hard · 4340 · Titanium Grade 5 · Inconel 718 · Aluminum 7075 · High-strength steels · Customer-specific alloys on request.

Defense parts routinely spec 4140 pre-hard, 4340, and precipitation-hardening stainless — condition-specific machining knowledge matters, same as aerospace. Titanium is common on weight-critical structural components. Legacy sustainment often specifies materials nobody sees anywhere else; we work with the spec sheet you provide and validate machining strategy before we cut.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

Tolerance and finish expectations in Military & Defense

±0.0005" routine on critical features. Tighter with proper setup and fixture. GD&T-driven tolerancing supported. CMM verification on first articles; customer-plan-driven measurement on production runs. Documentation the level a program office expects.

Documentation and quality workflow

ITAR-controlled work under NDA with customer-specific document control — prints stay in-shop. Customer-specific quality plans, PPAP, and first-article documentation standard. Material traceability by heat and lot on every job. Certificates of conformance with delivery.

Response time and emergency work

Program-driven schedules honored. Emergency component replacement supported for sustainment programs with established relationships. Legacy-part reverse-engineering supported for out-of-production sustainment.

How we compare

Compared to a large defense-prime captive shop: we're faster for one-offs, small runs, and legacy sustainment where the prime's supply chain has moved on. Compared to a lowest-bidder commodity shop: we handle ITAR, we document properly, and we understand that a program-office audit is a real thing.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request. ITAR-controlled prints handled under document control with prints staying in-shop.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Ready to talk about your Military & Defense work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

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